

DISEASE PREDICTION BASED ON SYMPTOMS

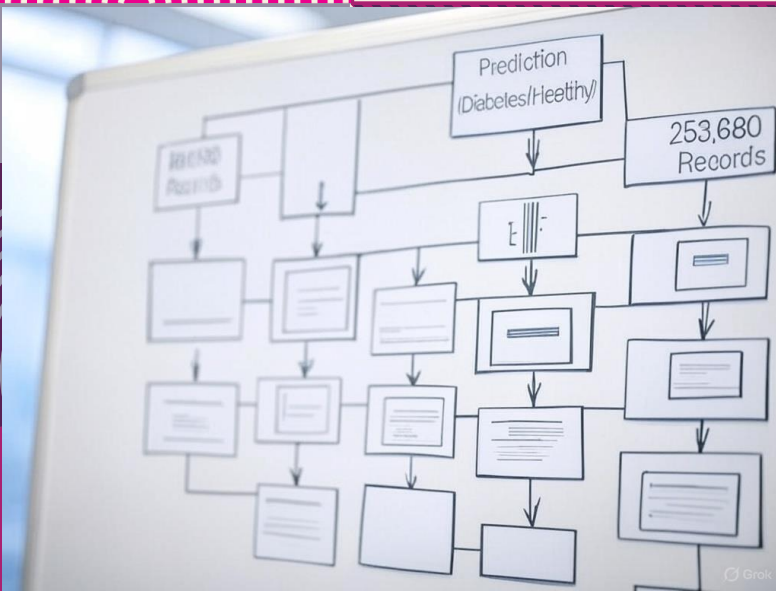
Leveraging Machine Learning for Diabetes and
Health Classification

Elliot Fisher
Mehdi Afiatpour
Sir James Jackson



DISEASE PREDICTION BASED ON SYMPTOMS

- Project Overview:
- Predict Diabetes and pre-diabetes with 21 health features (e.g., BMI, high BP, smoking, physical activity) from 253,680 records.
- Why It Matters: Early diagnosis improves outcomes; machine learning enhances accuracy and scalability in healthcare.



DATASET DESCRIPTION

Source: UCI Machine Learning Repository (Diabetes dataset) with 253,680 records.

Variables:
(21 features)

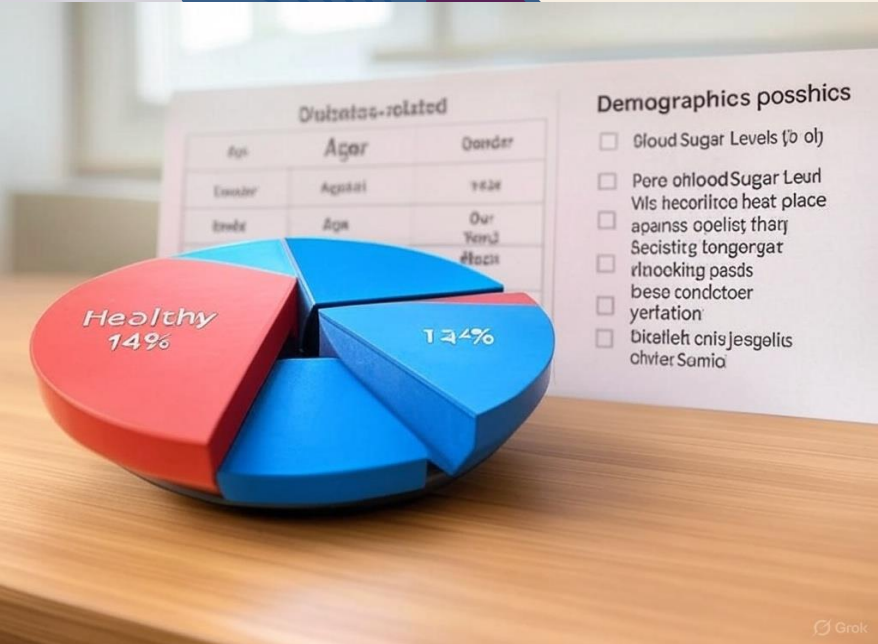
Demographics (e.g., Sex, Age)

Personal info (e.g., Education, Income)

Health history (e.g., High BP, Smoker, physical activity, BMI)

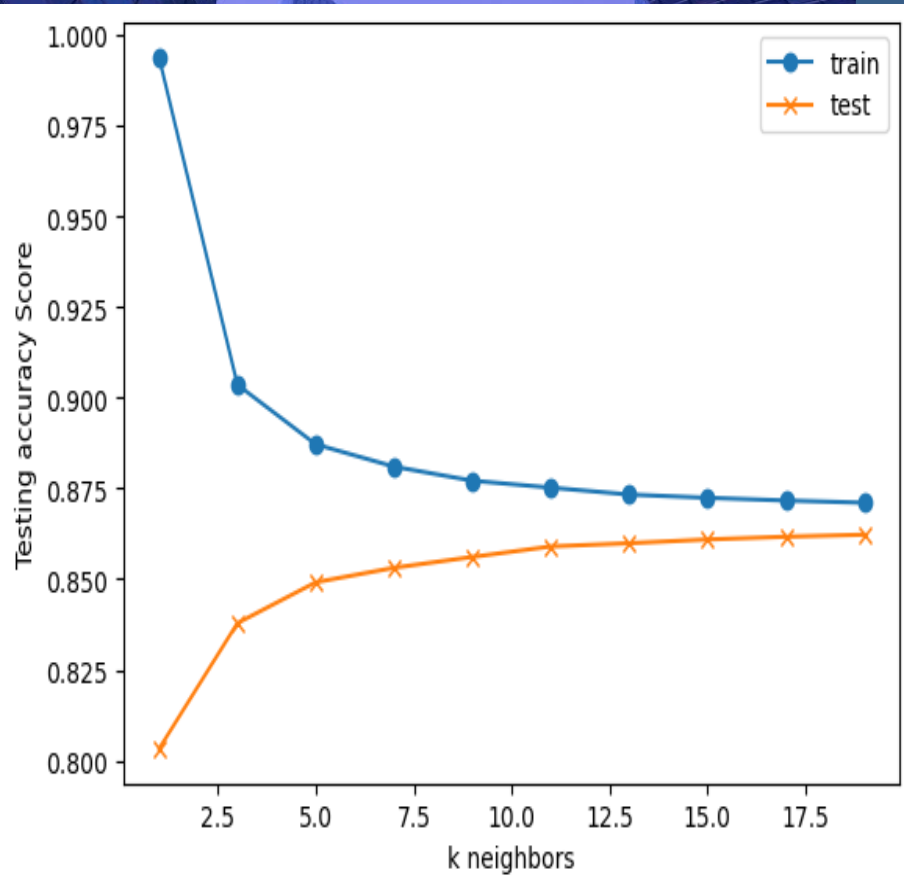
Class Labels:

Healthy (0), Diabetes/Pre-diabetes (1)



Category	Example Variabes	Type
Diabetes-related	High BP, BMI	Bool, Int
Demographics	Sex, Age	Bool, Int
Health history	Smoker, PhysActivity	Bool

SETTING UP GETTING STARTED



- Here's how we go from raw data to predictions.
- We import the UCI Diabetes dataset with `ucimlrepo` 253,680 records, 21 features like BMI and HighBP, binary targets (0 for Healthy, 1 for Diabetes/Pre-diabetes).
- Split it 70-30 for training and testing using `train_test_split` standard practice for model eval.
- Train a Random Forest Classifier as an example—it's one of several we used, like MLP and KNN. Fits the training data in one line.
- Classify with `predict` on the test set. RFC hit 86% accuracy, We'll dive into metrics later.
- This pipeline's quick and leverages sklearn, familiar stuff for you all.

MODEL EVALUATION

RFC accuracy: 0.86, precision (1): 0.50, recall (1): 0.17.

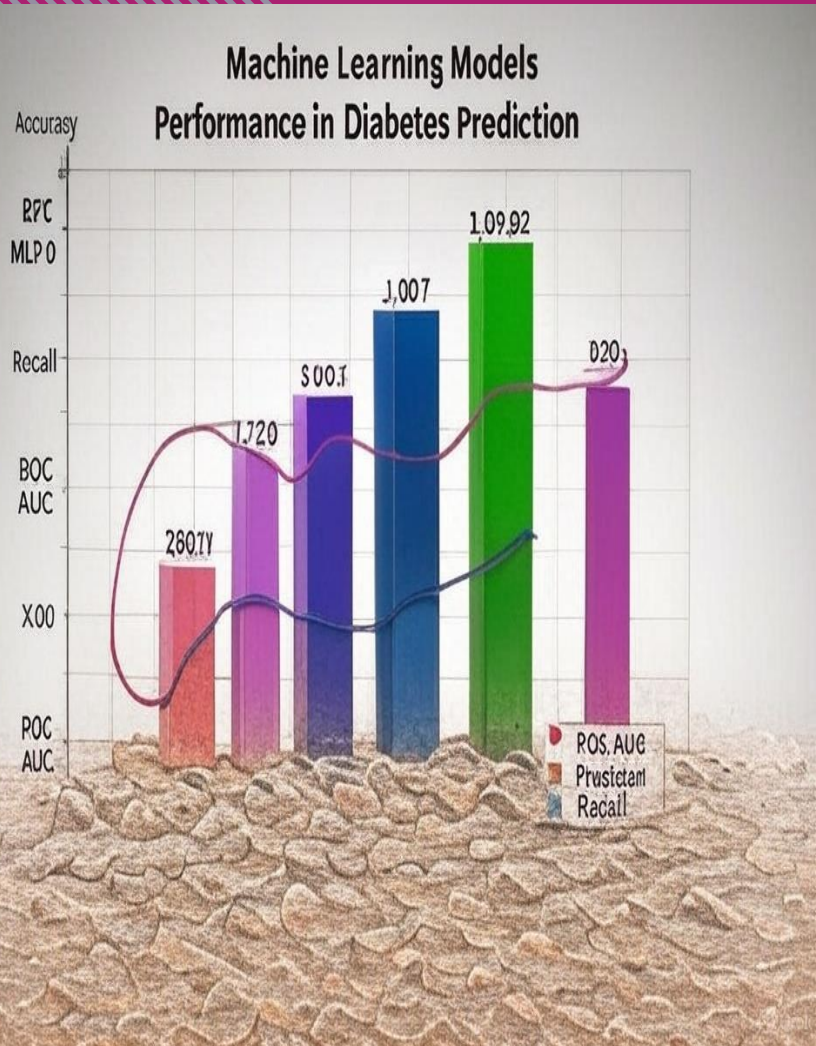
KNN accuracy: 0.85, precision (1): 0.43, recall (1): 0.22.

DT accuracy: 0.80

RFC's matrix ([[63561, 1789], [8958, 1796]]) visualizes performance, showing strengths (Healthy) and weaknesses (Diabetes)

	Predicted 0	Predicted 1
Actual 0	63531	1789
Actual 1	8958	1796

MODEL COMPARISON & SELECTION



- RFC: 86% accuracy, precision (1): 0.50, recall (1): 0.17, ROC
- AUC: 56.98%
- MLP: 86% accuracy, ROC AUC: 57.53%
- KNN: 85% accuracy, recall (1): 0.22
- DT: 80% accuracy, ROC AUC: 60.08%
- (ERT: 85%, LR: 86% omitted for simplicity, but can be added.)

MODEL DEPLOYMENT & USAGE

You may get the code from our github:

https://github.com/Mdfish9299/Project_Beta

Then simply run (in terminal):

Streamlit run app.py

Health Information

High Blood Pressure (0: No, 1: Yes): ☒ 0 ☐ 1	Heart Disease or Attack (0: No, 1: Yes): ☒ 0 ☐ 1	No Doctor because of cost in the past 12 months (0: No, 1: Yes): ☒ 0 ☐ 1	What is your Sex (0: Female, 1: Male): ☒ 0 ☐ 1
High Cholesterol (0: No, 1: Yes): ☒ 0 ☐ 1	Physical Activity in the past 30 days (0: No, 1: Yes): ☒ 0 ☐ 1	General Health Score: scale of 1-5 3 - +	What is your Age: 1 = 18-24, 9 = 60-64, 13 = 80 or older 4 - +
Cholesterol Check in 5 Years (0: No, 1: Yes): ☐ 0 ☒ 1	Fruits Consumption (0: No, 1: Yes): ☐ 0 ☒ 1	Mental Health Check: Days not good in the past 30 days 4 - +	What is your Level of Education: Scale 1-6 1 - +
BMI Score: 21 - +	Veggies Consumption (0: No, 1: Yes): ☒ 0 ☐ 1	Physical Health: Days not good in the past 30 days 2 - +	What is your Income: Scale 1-8 3 - +
Smoker (0: No, 1: Yes): ☒ 0 ☐ 1	Heavy Alcohol Consumption (0: No, 1: Yes): ☒ 0 ☐ 1	Difficulty Walking (0: No, 1: Yes): ☐ 0 ☒ 1	
Stroke (0: No, 1: Yes): ☒ 0 ☐ 1	Health Care Coverage (0: No, 1: Yes): ☒ 0 ☐ 1		

Make Prediction



THANK YOU

Elliot

Mehdi

Sir James